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NMAM INSTITUTE OF TECHNOLOGY, NITTE**Off-Campus Centre of Nitte (Deemed to be University)****Second Semester M.Tech. (VDE) (Credit System) Degree Examinations**

May – June 2025

22VDE222 - EMBEDDED CONTROLLER PROGRAMMING FOR REAL-TIME SYSTEMS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I**Marks BT* CO* PO***

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|-------|---|----|-----|---|---|
| 1. a) | Where do the temporal requirements come from? What are the parameters that describe the temporal characteristics of a controlled object? | 10 | L*2 | 1 | 3 |
| b) | What are the main differences between a soft real-time system and a hard real-time system? Explain them briefly. | 10 | L3 | 1 | 3 |
| 2. a) | For an embedded system, what are the distinctive characteristics that influence the system development process, and explain briefly each one of them? | 10 | L2 | 1 | 3 |
| b) | What is the fraction of development cost/production cost in embedded systems and in plant automation systems? How does this relation influence the system design? | 10 | L3 | 1 | 3 |
| 3. a) | With the help of any two examples explain why an end-to-end protocol is required at the interface between the computer system and the controlled object? | 10 | L2 | 2 | 3 |
| b) | With an example, illustrate the difference between Event-triggered and Time-Triggered real-time system. | 10 | L3 | 2 | 3 |

Unit – II

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|-------|---|----|----|---|---|
| 4. a) | Explain ARM memory organization and load store architecture. | 10 | L2 | 3 | 3 |
| b) | With a neat block diagram, explain the three-stage pipelining architecture used in ARM processor. | 10 | L2 | 3 | 3 |
| 5. a) | With a neat diagram, explain ARM's visible registers and also explain in brief the ARM XPSR format. | 10 | L2 | 3 | 3 |
| b) | With a neat block explain how Program memory and Data memory space allocation is done in the PIC18 microcontroller. | 10 | L3 | 4 | 3 |
| 6. a) | Explain the PIC18 microcontroller addressing modes with examples. | 10 | L2 | 4 | 3 |
| b) | With a neat block, explain how to read and write the PIC 18 Port structure. | 10 | L2 | 4 | 3 |

Unit – III

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|-------|---|----|----|---|---|
| 7. a) | Explain Timer 0 in 16-bit mode for PIC 18 microcontrollers with a neat diagram. Give a brief explanation of the registers involved. | 10 | L2 | 5 | 3 |
| b) | Explain the block diagram of the PIC18F452 A/D converter and explain the registers involved in the conversion. | 10 | L2 | 5 | 3 |

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8. a) Explain the core and peripheral interrupt sources available in the PIC18 microcontroller; also mention the ten registers that control interrupt operations.
- b) Explain in brief the registers of the PIC18 UART required while programming the controller.

10	L2	5	3
10	L2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
